Auditing AI Output

Data Can Deceive · Causal Roadmaps with (and without) AI

Module E

August 5, 2026

Auriane Journet

A Confident Answer Is Not a Correct One

- The model will **not** tell you when it is wrong; fluency is not validity.
- "Human in the loop" does *not* mean reviewing the output and nodding.
- It means **auditing each claim** against the evidence that would justify it.

Our test case

An **adjustment set** for a **target-trial protocol** [1] — the list of variables an analysis will control for.

Objective: turn a confident list of covariates into a set of checkable claims.

Test Case

Our prompt

"Among adults initiating an SGLT2 inhibitor versus a DPP-4 inhibitor for type 2 diabetes, does the SGLT2 inhibitor reduce hospitalization for heart failure? List the variables to adjust for so the groups are comparable."

The model's answer

"Adjust for: age, sex, baseline eGFR, baseline HbA1c, prior heart failure, prior MI, baseline diuretic use, HbA1c at 6 months, and change in eGFR on treatment. Adjusting for glycemic control and renal response removes any residual imbalance in disease severity."

Side note: models vary and change fast, a good answer today can be a bad one tomorrow. The key point is to build the habit of auditing a model's answer *regardless* of how confident it sounds.

Step 1: First Pass

Answer in the chat

Read the nine variables suggested by the model. **Flag any you would question** and say why in a few words.

- age
- sex
- baseline eGFR
- baseline HbA1c
- prior heart failure
- prior MI
- baseline diuretic use
- HbA1c at 6 months
- change in eGFR on treatment

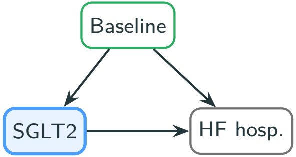
Step 2: Audit

- Baseline timing is **necessary but not sufficient**: accept a covariate only when substantive knowledge and the DAG support a common-cause role.
- HbA1c at 6 months and on-treatment eGFR change are measured **after** treatment starts.
- They sit on the causal pathway (**mediators**) and can act as **colliders**.
- Adjusting for them **blocks part of the effect** you want and can **manufacture bias**.

Baseline vs Post-Baseline

Baseline confounder

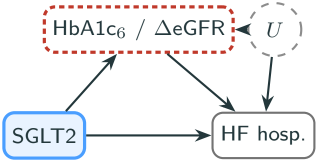
measured *before* time zero



Adjust to close the backdoor.

Post-baseline variable

HbA1c_6 / ΔeGFR, measured *after* time zero



Do not adjust — removes the *mediated part* of the total effect *and* opens collider bias via U.

Assumption: the collider path needs $U \to M$ (unmeasured severity affects on-treatment response).

Drawing out the DAG helps make these assumptions clearer.

Step 3: Decide and Justify

- Every field gets one of three verdicts: **ACCEPT** / **CORRECT** / **REJECT**.
- Each verdict is **paired with the evidence** that justifies it (otherwise it is just another confident assertion).
- For each covariate in your set, answer:
  - Was it measured **before time zero**?
  - **Where does it sit on the DAG** (confounder, mediator, collider)?

Next step

ACCEPT keep as-is **CORRECT** keep but fix (definition / timing / add a missing one)
REJECT remove

Step 4: Keep an Audit Log

Variable	Timing	Verdict	Justification
age, sex	pre-time-zero	**ACCEPT**	Fixed at baseline; affect prescribing and HF risk.
baseline eGFR, HbA1c	pre-time-zero	**ACCEPT**	Indication / severity confounders present before treatment.
prior HF, prior MI	pre-time-zero	**ACCEPT**	Strong confounders of treatment choice and outcome.
baseline diuretic use	pre-time-zero	**CORRECT**	Keep, but flag as a *proxy* for HF severity; define the lookback window.
HbA1c at 6 months	post-baseline	**REJECT**	Mediator on $A \to Y$; collider with U. Adjusting biases the total effect.
ΔeGFR on treatment	post-baseline	**REJECT**	On the causal pathway; conditioning removes the mediated part and opens a collider.

When the Mediator Is the Question

ACCEPT / **REJECT** assume you want the **total effect** of treatment. For that estimand, "post-baseline $\Rightarrow$ reject" is right.

Controlling for mediators

If the *mediated pathway itself* is your question, how much of the effect runs *through* HbA1c or eGFR, that is a **mediation analysis**. With treatment-affected confounding it calls for **g-methods** (g-computation, IPTW, TMLE) [2].

Reject the mediator *for a total effect*, and model it deliberately when it *is* the target. Do not apply the checklist bluntly.

Your Turn!

Target trial. Among adults with COPD initiating **inhaled corticosteroids (ICS)** versus a long-acting bronchodilator alone, do ICS reduce **severe exacerbations** over the next year?

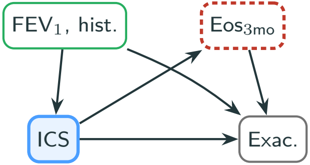
Proposed adjustment set:

- age
- prior-year exacerbation count
- baseline blood eosinophils
- smoking
- blood eosinophils at 3 months on ICS

Your task: for each variable, draw a DAG and run the audit test $\Rightarrow$ **ACCEPT** / **CORRECT** / **REJECT**. **Anything missing?**

Solution

- age, prior exacerbations, baseline eosinophils: **ACCEPT**
  pre-time-zero confounders of ICS use and exacerbations
- smoking: **CORRECT**
  keep, but define: baseline status / pack-years, before time zero
- eosinophils at 3 months: **REJECT**
  post-baseline mediator: ICS lowers eosinophils $\rightarrow$ fewer exacerbations
- *missing* baseline FEV_1: **CORRECT** (**add**)
  airflow severity confounds both ICS use and exacerbations



green = adjust red box = do not

Takeaways

Auditing any adjustment set

1. **Question the model's output**: is each variable measured *before* treatment starts? What is a given variable's position on a DAG (confounder, mediator, or collider)?
2. **Audit**: **ACCEPT** / **CORRECT** / **REJECT** for each. An omitted confounder is a finding too (**CORRECT** $\rightarrow$ add).
3. **Justify**
4. **Document**

The completed log *is* your causal assumptions made explicit and reproducible, as opposed to a black-box list of covariates [2].

Audit each claim against its evidence.

Module F carries the corrected roadmap into the Navigator.

References

[1] Hernán MA, Robins JM. Using big data to emulate a target trial when a randomized trial is not available. *American Journal of Epidemiology*, 183(8):758–764, 2016.

[2] Hernán MA, Robins JM. *Causal Inference: What If*. Chapman & Hall/CRC, 2020.